Addressing Questions/Comments received for DY absolute Cross-Section Study

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Overview

This talk addresses additional comments received for the talk DocID: 11584
(https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11584)

Supplemental Documents:

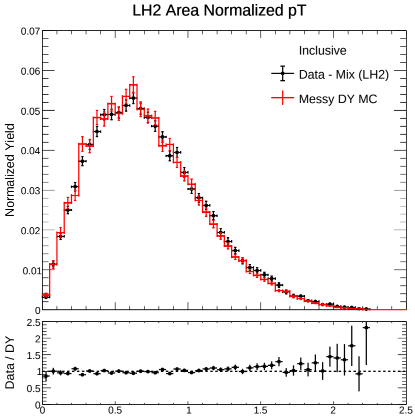
- Technote
  (https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11569)
- Previous talk
  (https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11584)

Additional Data/MC plots

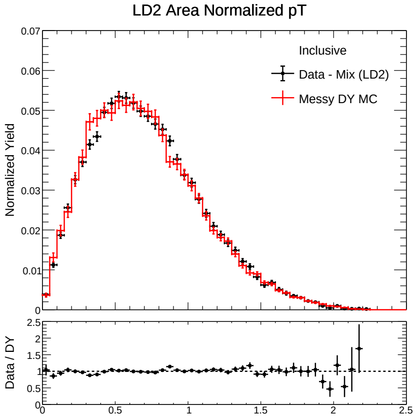
"The other comparison that I raised on Friday was of the mass and xF distributions in each pT bin. Can you make comparison plots? As you integrate the events over mass and xF when deriving dsigma/dpT, the agreement between the real data and the MC is crucial"

p_T Inclusive (Data - Mix): LH2 vs LD2

LH2 Target

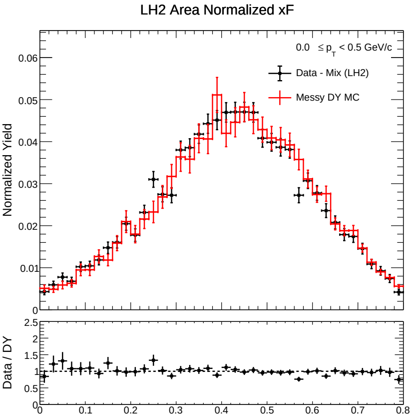


LD2 Target

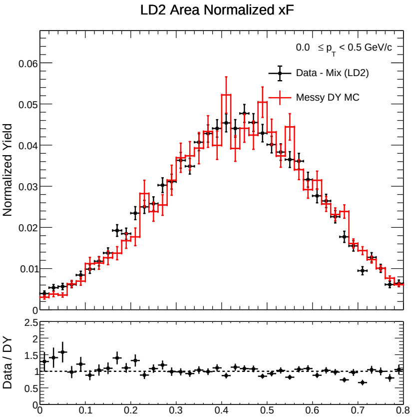


x_F $(0.0 \le p_T < 0.5$ GeV/c): LH2 vs LD2

LH2 Target

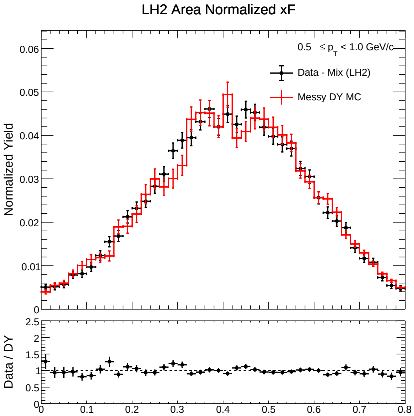


LD2 Target

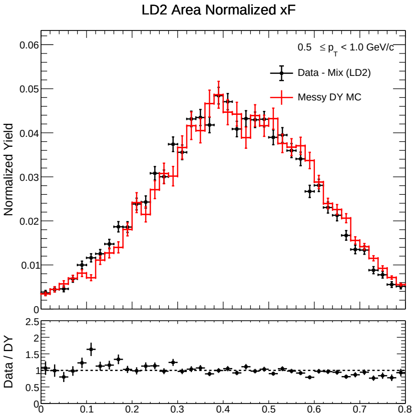


x_F $(0.5 \leq p_T < 1.0$ GeV/c): LH2 vs LD2

LH2 Target

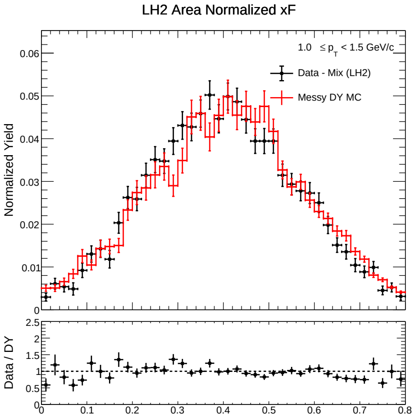


LD2 Target

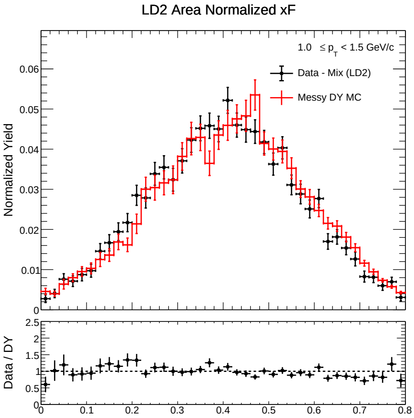


x_F $(1.0 \leq p_T < 1.5$ GeV/c): LH2 vs LD2

LH2 Target

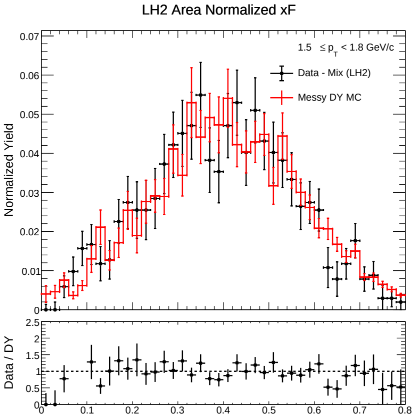


LD2 Target

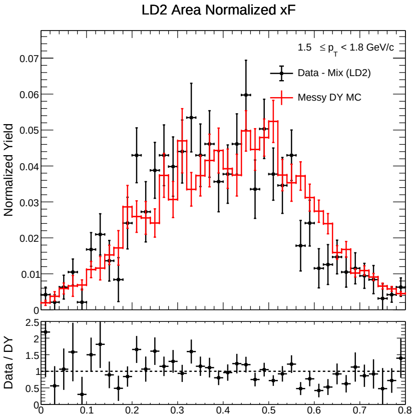


x_F $(1.5 \leq p_T < 1.8$ GeV/c): LH2 vs LD2

LH2 Target

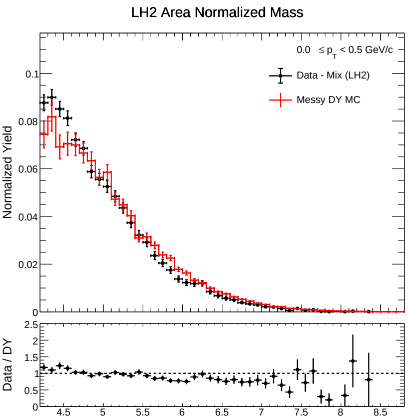


LD2 Target

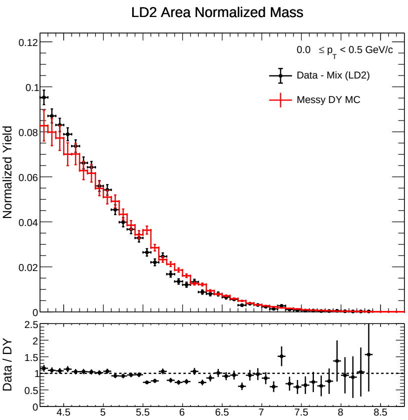


Mass ($0.0 \leq p_T < 0.5$ GeV/c): LH2 vs LD2

LH2 Target

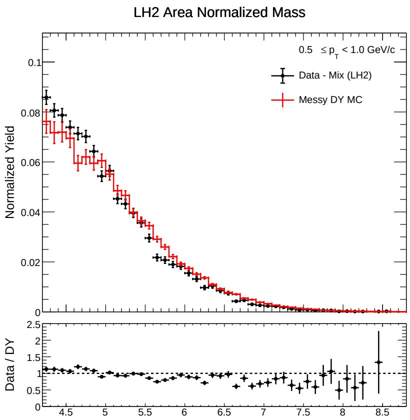


LD2 Target

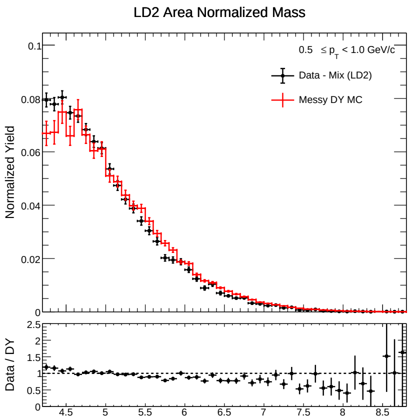


Mass ($0.5 \leq p_T < 1.0$ GeV/c): LH2 vs LD2

LH2 Target

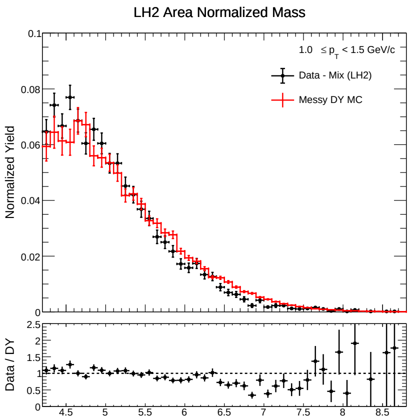


LD2 Target

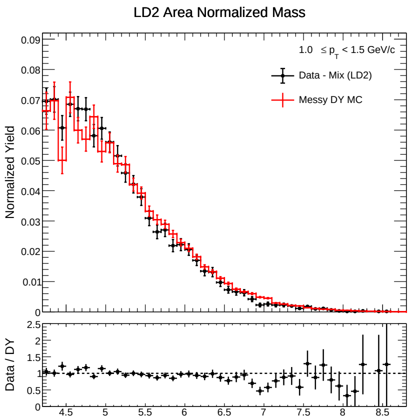


Mass ($1.0 \leq p_T < 1.5$ GeV/c): LH2 vs LD2

LH2 Target

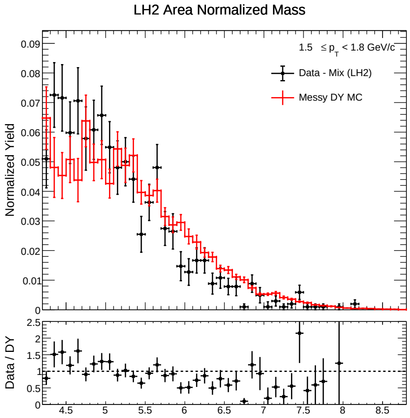


LD2 Target

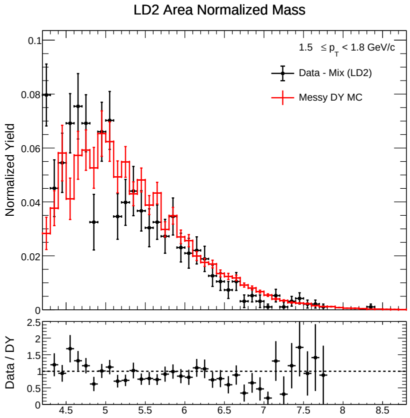


Mass ($1.5 \leq p_T < 1.8$ GeV/c): LH2 vs LD2

LH2 Target



LD2 Target



Systematic Correlations to cross-section ratio

"How was the systematic error propagated from the LH2 and LD2 results? Have you considered any correlation?"

The Observable: Cross-Section Ratio

Objective: Calculate the error for the single differential cross-section ratio in a given kinematic bin (e.g., p_T).

The ratio R is defined as:

$$R = \frac{\sigma_{pd}}{2\sigma_{pp}} \equiv \frac{A}{cB} \tag{1}$$

Where:

- $A = \sigma_{pd}$ (Liquid Deuterium cross-section)
- $B = \sigma_{pp}$ (Liquid Hydrogen cross-section)
- $c = 2$ (Exact scaling constant, zero uncertainty)

General Error Propagation (The Master Equation)

For any function $R(A, B)$, the variance is given by the partial derivatives and the covariance matrix:

$$\delta R^2 = \left(\frac{\partial R}{\partial A}\right)^2 \delta A^2 + \left(\frac{\partial R}{\partial B}\right)^2 \delta B^2 + 2\left(\frac{\partial R}{\partial A}\right)\left(\frac{\partial R}{\partial B}\right) \mathrm{cov}(A, B) \tag{2}$$

By defining the covariance in terms of the Pearson correlation coefficient ρ:

$$\mathrm{cov}(A, B) = \rho(\delta A)(\delta B) \quad \text{where } \rho \in [-1, 1] \tag{3}$$

Evaluating the derivatives for $R = A/(cB)$ and converting to relative errors yields the master equation:

$$\left(\frac{\delta R}{R}\right)^2 = \left(\frac{\delta A}{A}\right)^2 + \left(\frac{\delta B}{B}\right)^2 - 2\rho\left(\frac{\delta A}{A}\right)\left(\frac{\delta B}{B}\right) \tag{4}$$

1. Statistical Uncertainties ($\rho = 0$)

Physical Reality: Statistical fluctuations are driven by random counting statistics in finite data samples.

A random upward fluctuation in the *pd* dataset has no influence on the random fluctuations in the *pp* dataset. Therefore, they are **completely uncorrelated** ($\rho = 0$).

Substituting $\rho = 0$ into the master equation:

$$\left(\frac{\delta R_{\text{stat}}}{R}\right)^2 = \left(\frac{\delta\sigma_{pd}^{\text{stat}}}{\sigma_{pd}}\right)^2 + \left(\frac{\delta\sigma_{pp}^{\text{stat}}}{\sigma_{pp}}\right)^2 - 2(0)\left(\frac{\delta\sigma_{pd}^{\text{stat}}}{\sigma_{pd}}\right)\left(\frac{\delta\sigma_{pp}^{\text{stat}}}{\sigma_{pp}}\right)$$

The covariance term vanishes, yielding the quadrature sum:

$$\frac{\delta R_{\text{stat}}}{R} = \sqrt{\left(\frac{\delta\sigma_{pd}^{\text{stat}}}{\sigma_{pd}}\right)^2 + \left(\frac{\delta\sigma_{pp}^{\text{stat}}}{\sigma_{pp}}\right)^2} \tag{5}$$

2. Systematic Uncertainties ($\rho = 1$)

Physical Reality: Systematic biases (e.g., tracking efficiency, geometric acceptance) apply to the exact same spectrometer running under identical beam conditions.

If an inefficiency causes us to systematically underestimate σ_{pd}, it will also cause us to systematically underestimate σ_{pp}. They move together (**perfectly positively correlated**, $\rho = 1$).

Substituting $\rho = 1$ into the master equation:

$$\left(\frac{\delta R_{\text{sys}}}{R}\right)^2 = \left(\frac{\delta\sigma_{pd}^{\text{sys}}}{\sigma_{pd}}\right)^2 + \left(\frac{\delta\sigma_{pp}^{\text{sys}}}{\sigma_{pp}}\right)^2 - {\color{red}2(1)}\left(\frac{\delta\sigma_{pd}^{\text{sys}}}{\sigma_{pd}}\right)\left(\frac{\delta\sigma_{pp}^{\text{sys}}}{\sigma_{pp}}\right)$$

This forms a perfect square $x^2 + y^2 - 2xy = (x - y)^2$:

$$\left(\frac{\delta R_{\text{sys}}}{R}\right)^2 = \left(\frac{\delta\sigma_{pd}^{\text{sys}}}{\sigma_{pd}} - \frac{\delta\sigma_{pp}^{\text{sys}}}{\sigma_{pp}}\right)^2 \tag{6}$$

The Power of Ratios: Error Cancellation

Taking the square root gives our final systematic error propagation:

$$\frac{\delta R_{\text{sys}}}{R} = \left| \frac{\delta\sigma_{pd}^{\text{sys}}}{\sigma_{pd}} - \frac{\delta\sigma_{pp}^{\text{sys}}}{\sigma_{pp}} \right| \tag{7}$$

Why this matters:

- If systematic errors are evaluated as uncorrelated, they inflate the final uncertainty (quadrature sum).
- By correctly identifying them as perfectly correlated ($\rho = 1$), identical fractional biases subtract and **cancel out entirely**.
- This mathematically isolates the physical cross-section ratio from global detector biases.

Total Bin-by-Bin Uncertainty

Finally, we evaluate the total error for each kinematic bin.

Because the sources of statistical scatter (random sample size) and systematic bias (global normalization/efficiencies) are independent of each other, their cross-correlation is zero.

We add the absolute propagated errors in quadrature:

$$\delta R_{\text{total}} = \sqrt{(\delta R_{\text{stat}})^2 + (\delta R_{\text{sys}})^2} \tag{8}$$

This δR_{total} represents the final error bars plotted on the single differential cross-section ratio $d\sigma_{pd}/2d\sigma_{pp}$ vs p_T.

Error Table Comparison

Summary Table: Extracted bin uncertainties comparing the uncorrelated background treatment against the new 100% correlated systematic treatment.

Uncorrelated (Old)

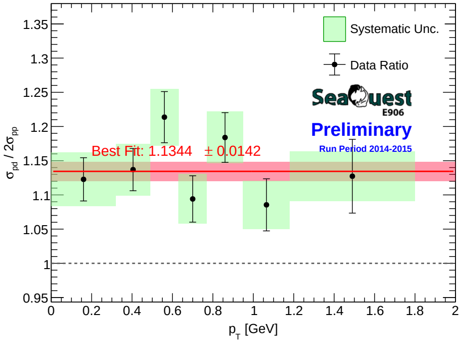
p_T Bin [GeV]	Ratio	Stat Err	Sys Err	Total Err
[0.00, 0.32)	1.1228	0.0316	0.0392	0.0504
[0.32, 0.49)	1.1370	0.0309	0.0376	0.0487
[0.49, 0.63)	1.2137	0.0374	0.0413	0.0557
[0.63, 0.77)	1.0941	0.0339	0.0365	0.0498
[0.77, 0.95)	1.1839	0.0364	0.0377	0.0524
[0.95, 1.18)	1.0855	0.0381	0.0351	0.0518
[1.18, 1.80)	1.1273	0.0539	0.0362	0.0649

Correlated (Latest)

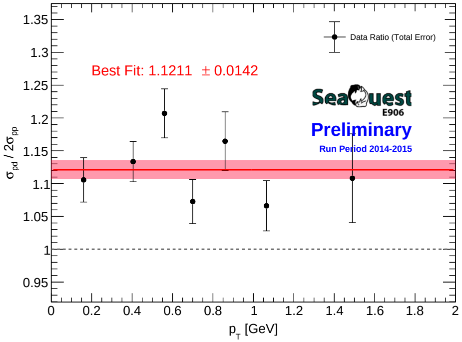
p_T Bin [GeV]	Ratio	Stat Err	Sys Err	Total Err
[0.00, 0.32)	1.1057	0.0311	0.0130	0.0338
[0.32, 0.49)	1.1335	0.0308	0.0006	0.0308
[0.49, 0.63)	1.2070	0.0372	0.0037	0.0374
[0.63, 0.77)	1.0727	0.0332	0.0056	0.0337
[0.77, 0.95)	1.1646	0.0358	0.0269	0.0448
[0.95, 1.18)	1.0662	0.0374	0.0080	0.0383
[1.18, 1.80)	1.1081	0.0530	0.0421	0.0677

Ratio Plot Comparison

Uncorrelated Background Treatment (Old)



100% Correlated Propagation (Latest)



Weighted Average Cross-Section

"Can you compute the weighted standard deviation of the cross sections of RS 57, 59, 62, 67 and 70? It should be better in the sense that we treat the roadsets equally, since we cannot assure RS 67 is 100% correct."

This requires collaboration input!

Variance Calculation Mathematics

For a given kinematic bin, let x_i be the absolute cross-section for roadset i, and e_i be its statistical error.

Inverse-Variance Weighting:
To suppress massive Poisson fluctuations from low-POT datasets (e.g., RS59), we scale by statistical power ($w_i = 1/e_i^2$):

$$\mu_w = \frac{\sum w_i x_i}{\sum w_i}$$

$$\sigma_{sys,w} = \sqrt{\frac{\sum w_i}{\left(\sum w_i\right)^2 - \sum w_i^2} \sum w_i (x_i - \mu_w)^2}$$

Resulting Uncertainties (1D p_T - LH2)

Using the inverse-variance weighted strategy suppresses low-statistics noise, yielding the following systematic errors for Liquid Hydrogen.
Note: Bins with zero relative error or relative error $\geq$ 50% have been omitted.

p_T **Bin**	**Range [GeV]**	**Mean** σ_w	**Std Dev** $\sigma_{sys,w}$	**Relative Error**
0	$[0.00, 0.32)$	0.001794	0.000290	16.18%
1	$[0.32, 0.49)$	0.003872	0.000511	13.19%
2	$[0.49, 0.63)$	0.004383	0.000536	12.24%
3	$[0.63, 0.77)$	0.004766	0.000651	13.65%
4	$[0.77, 0.95)$	0.004313	0.000533	12.37%
5	$[0.95, 1.18)$	0.003721	0.000662	17.79%
6	$[1.18, 1.80)$	0.001853	0.000248	13.40%

Resulting Uncertainties (1D p_T - LD2)

Applying the same inverse-variance weighted procedure to Liquid Deuterium targets yields comparable systematic margins.

Note: Bins with zero relative error or relative error $\geq$ 50% have been omitted.

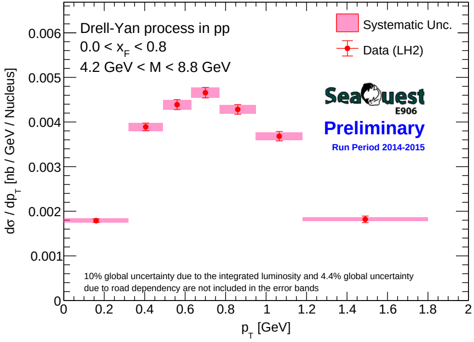
p_T **Bin**	**Range [GeV]**	**Mean** σ_w	**Std Dev** $\sigma_{sys,w}$	**Relative Error**
0	$[0.00, 0.32)$	0.003968	0.000698	17.59%
1	$[0.32, 0.49)$	0.008777	0.001154	13.15%
2	$[0.49, 0.63)$	0.010581	0.001340	12.67%
3	$[0.63, 0.77)$	0.010224	0.001331	13.02%
4	$[0.77, 0.95)$	0.010045	0.001534	15.27%
5	$[0.95, 1.18)$	0.007935	0.001482	18.67%
6	$[1.18, 1.80)$	0.004107	0.000738	17.98%

List of systematics

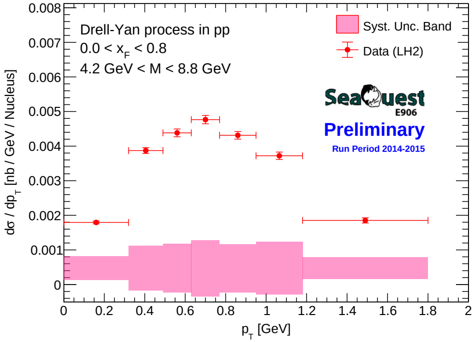
- Acceptance Correction
- Efficiency Correction for Yields
- ψ' Contamination
- Road Dependancy

LH2 Cross-Section Previous Vs Latest with updated systematics

Previous

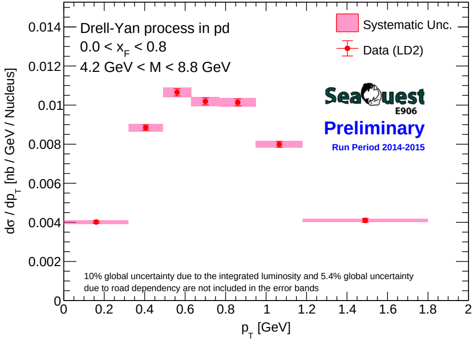


Latest (with weighted average)

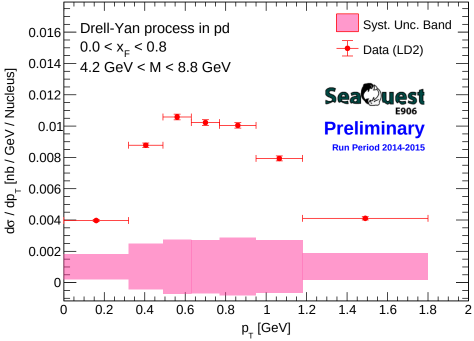


LD2 Cross-Section Previous Vs Latest with updated systematics

Previous

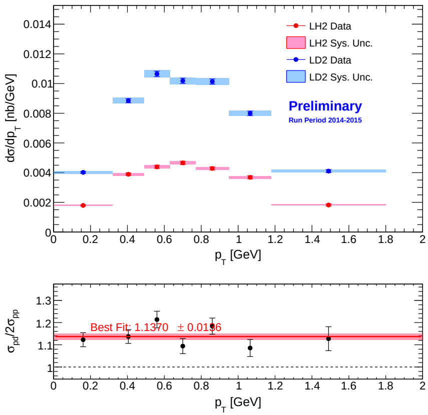


Latest (with weighted average)

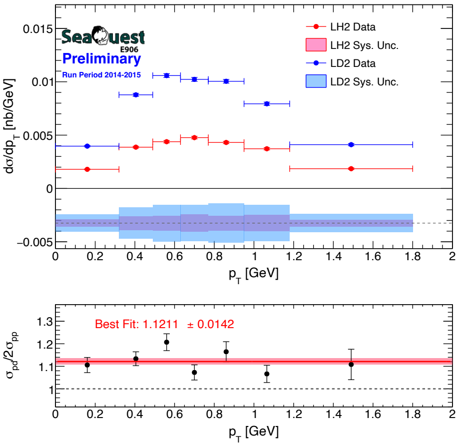


Summary Plots for p_T Previous Vs Latest with updated systematics

Previous

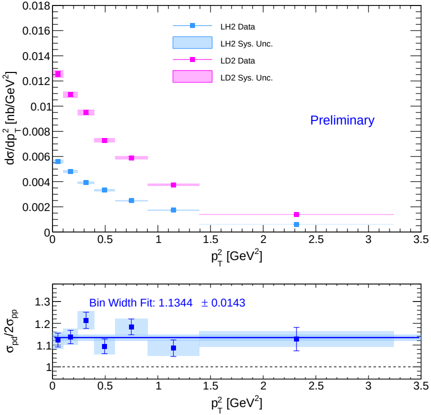


Latest (with weighted average)

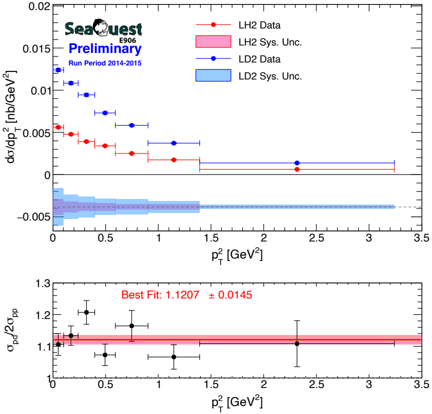


Summary Plots for p_T^2 Previous Vs Latest with updated systematics

Previous

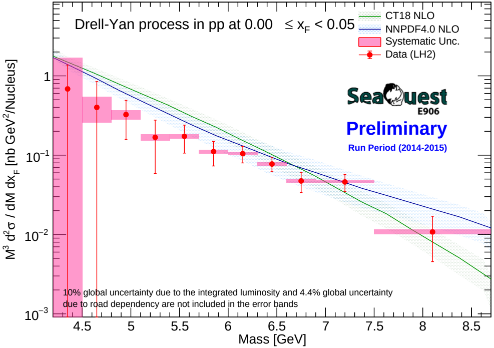


Latest (with weighted average)

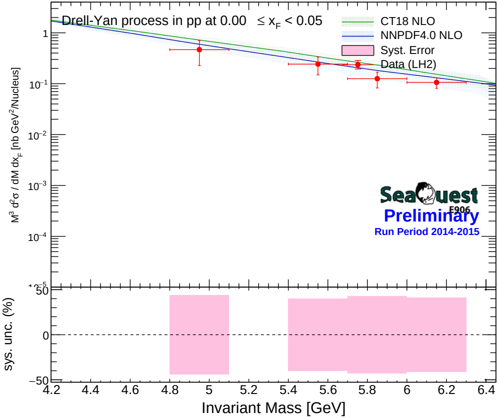


LH2 Cross-Section Previous Vs Latest: $0.00 \leq x_F < 0.05$

Previous

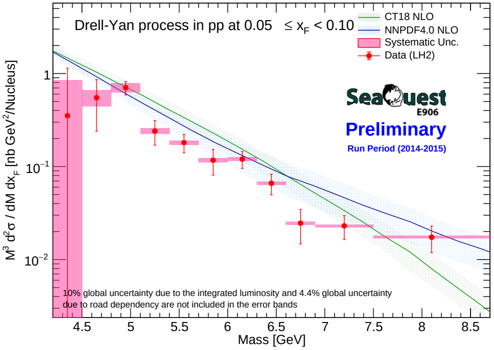


Latest (with weighted average)

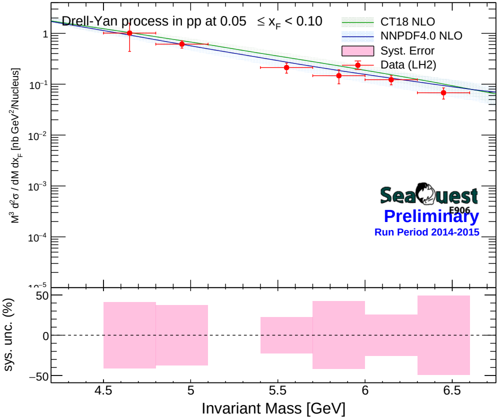


LH2 Cross-Section Previous Vs Latest: $0.05 \leq x_F < 0.10$

Previous

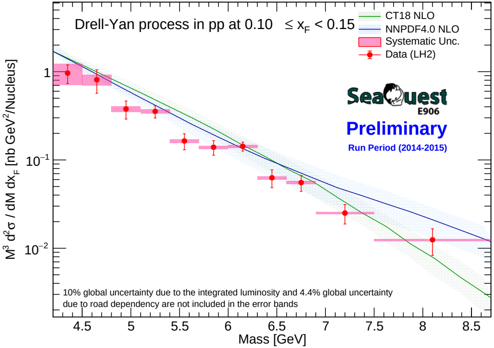


Latest (with weighted average)

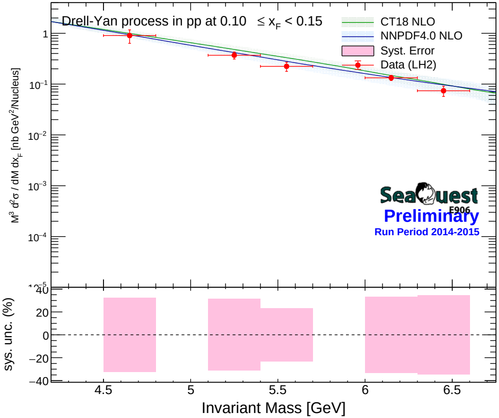


LH2 Cross-Section Previous Vs Latest: $0.10 \leq x_F < 0.15$

Previous

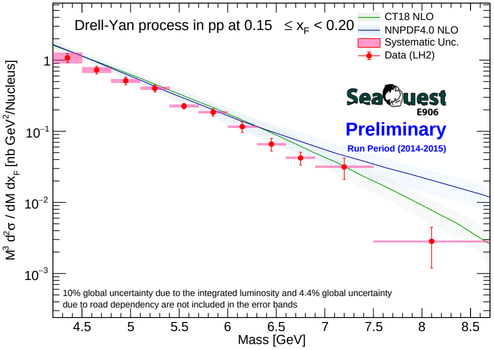


Latest (with weighted average)

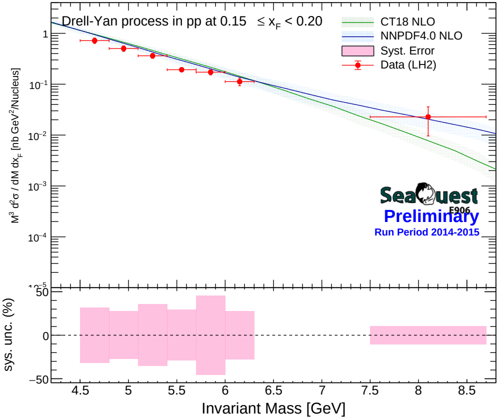


LH2 Cross-Section Previous Vs Latest: $0.15 \leq x_F < 0.20$

Previous

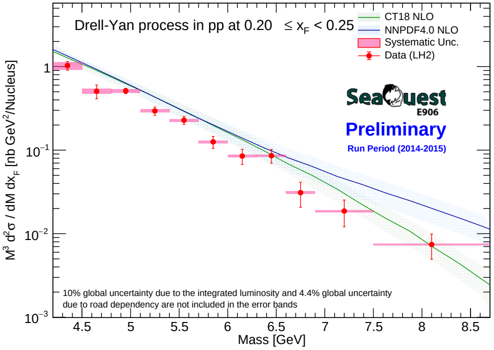


Latest (with weighted average)

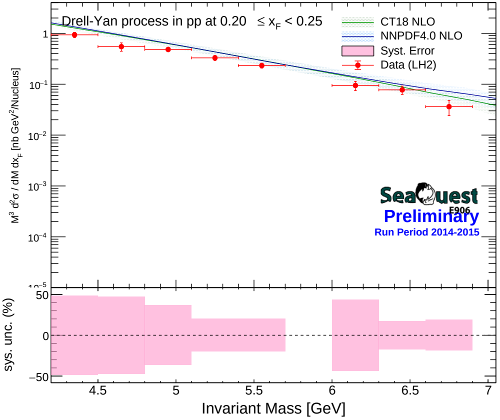


LH2 Cross-Section Previous Vs Latest: $0.20 \leq x_F < 0.25$

Previous

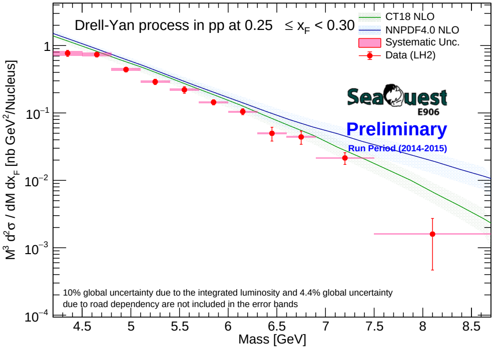


Latest (with weighted average)

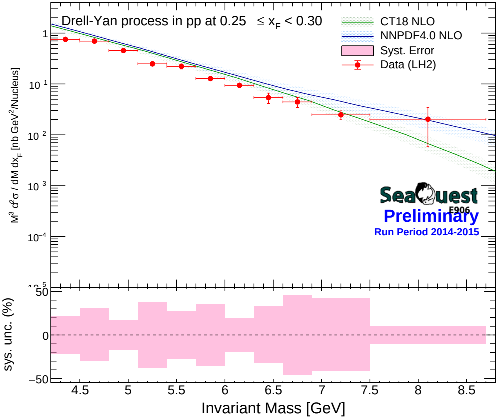


LH2 Cross-Section Previous Vs Latest: $0.25 \leq x_F < 0.30$

Previous

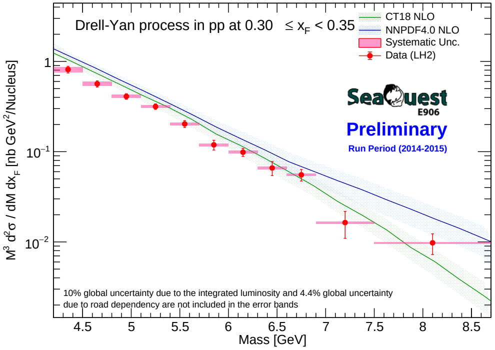


Latest (with weighted average)

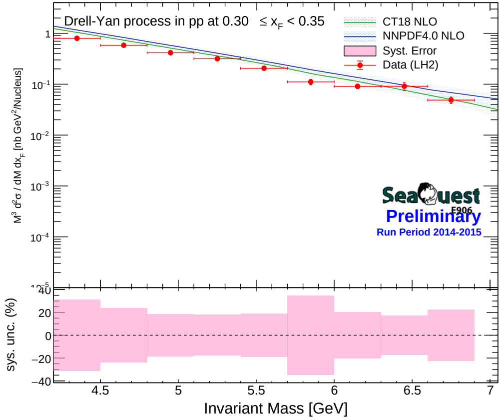


LH2 Cross-Section Previous Vs Latest: $0.30 \leq x_F < 0.35$

Previous

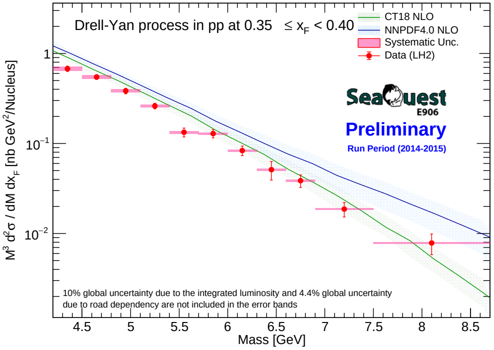


Latest (with weighted average)

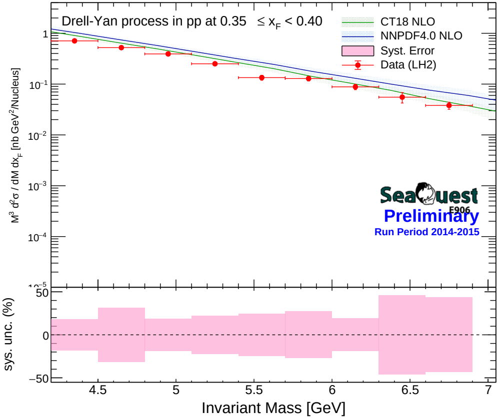


LH2 Cross-Section Previous Vs Latest: $0.35 \leq x_F < 0.40$

Previous

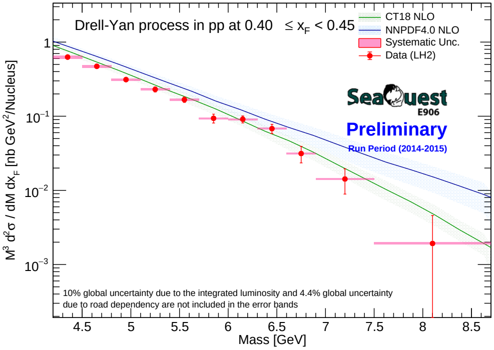


Latest (with weighted average)

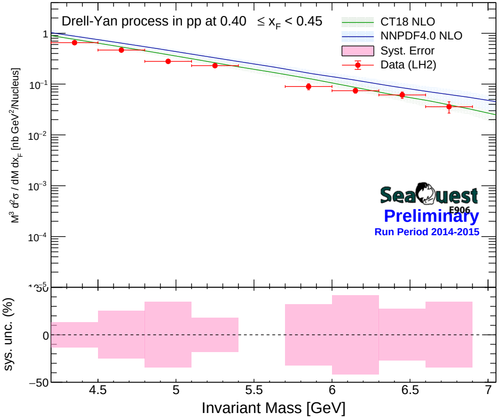


LH2 Cross-Section Previous Vs Latest: $0.40 \leq x_F < 0.45$

Previous

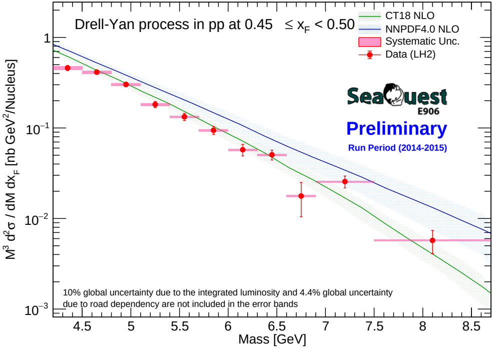


Latest (with weighted average)

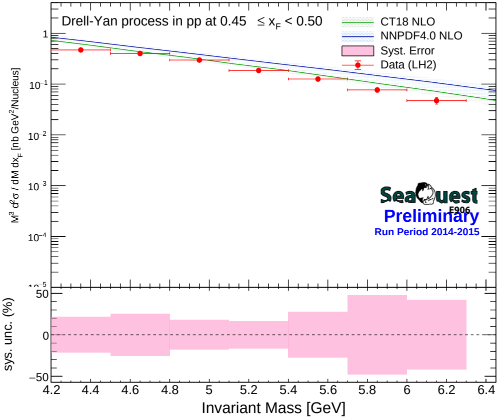


LH2 Cross-Section Previous Vs Latest: $0.45 \leq x_F < 0.50$

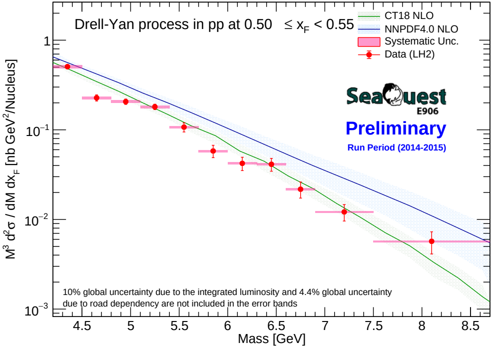
Previous



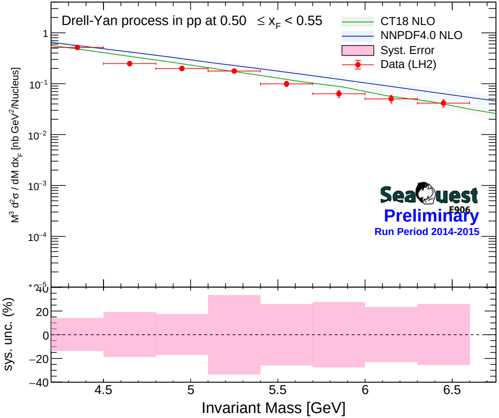
Latest (with weighted average)



Previous

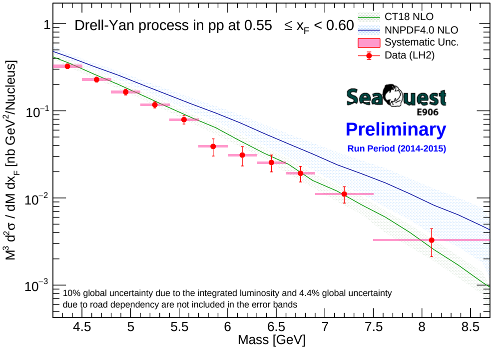


Latest (with weighted average)

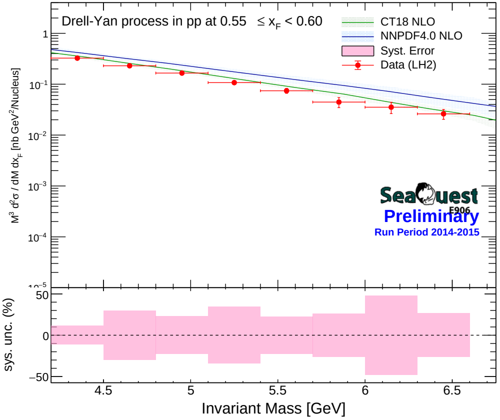


LH2 Cross-Section Previous Vs Latest: $0.55 \leq x_F < 0.60$

Previous

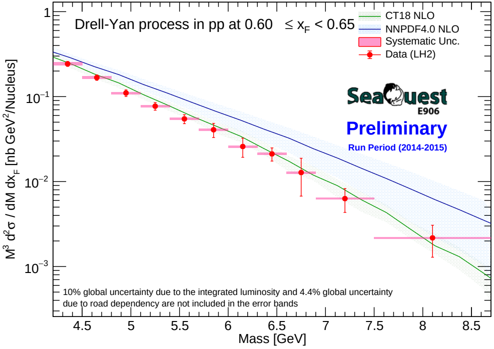


Latest (with weighted average)

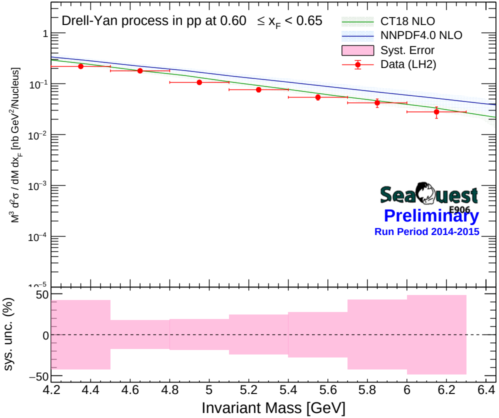


LH2 Cross-Section Previous Vs Latest: $0.60 \le x_F < 0.65$

Previous

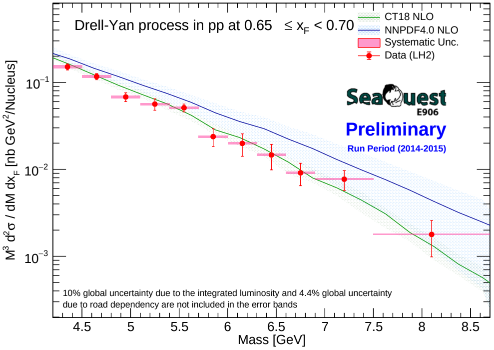


Latest (with weighted average)

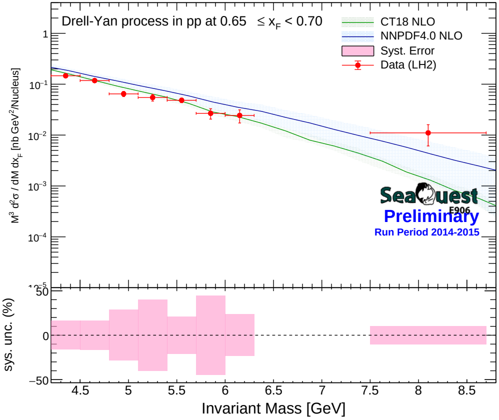


LH2 Cross-Section Previous Vs Latest: $0.65 \leq x_F < 0.70$

Previous

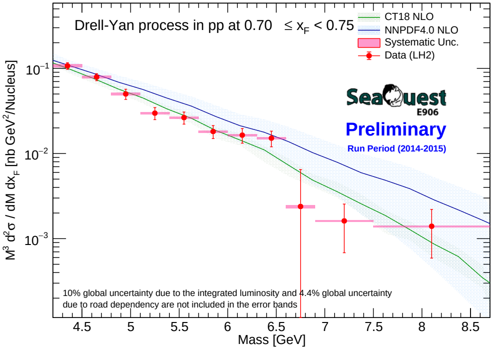


Latest (with weighted average)

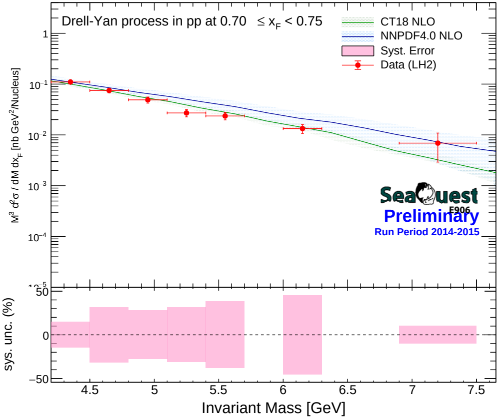


LH2 Cross-Section Previous Vs Latest: $0.70 \leq x_F < 0.75$

Previous

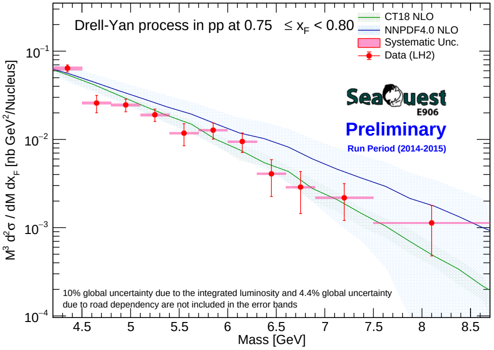


Latest (with weighted average)

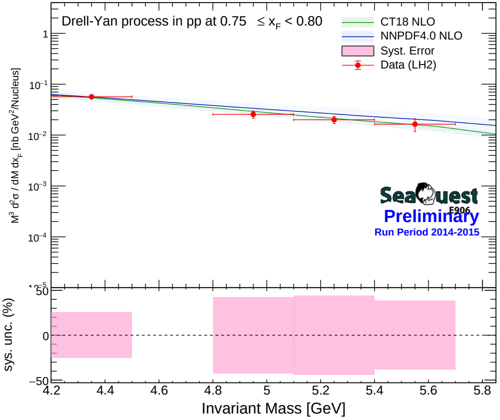


LH2 Cross-Section Previous Vs Latest: $0.75 \leq x_F < 0.80$

Previous

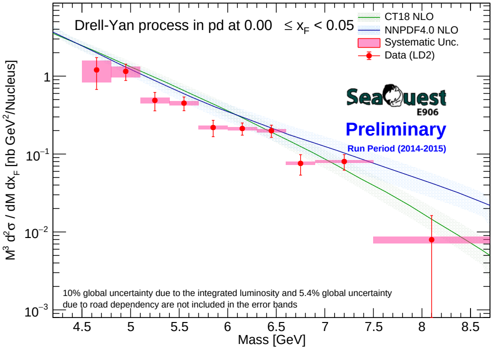


Latest (with weighted average)

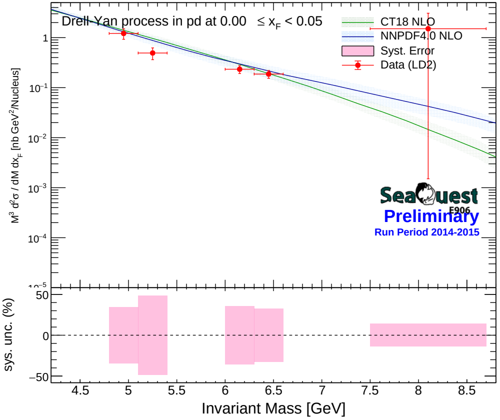


LD2 Cross-Section Previous Vs Latest: $0.00 \leq x_F < 0.05$

Previous

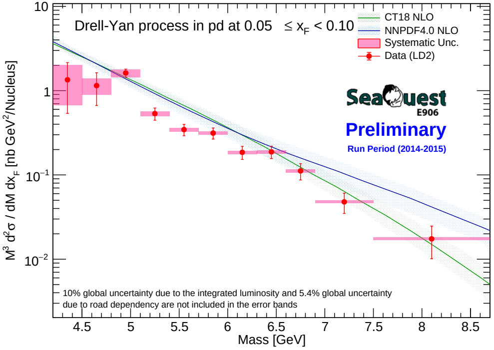


Latest (with weighted average)

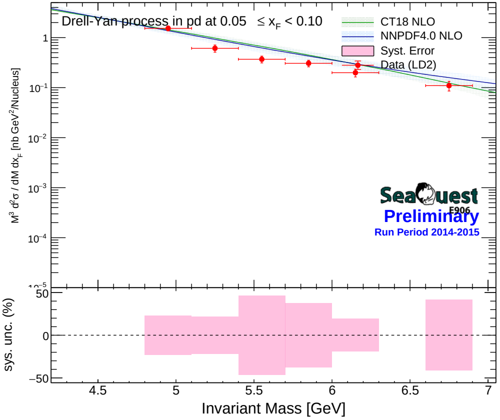


LD2 Cross-Section Previous Vs Latest: $0.05 \leq x_F < 0.10$

Previous

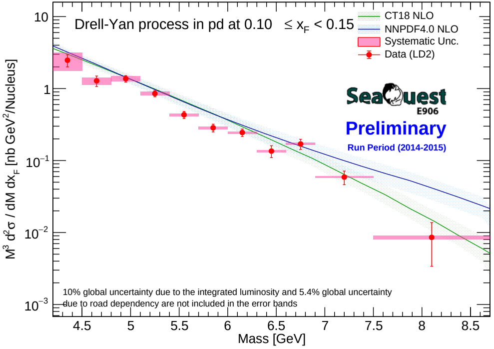


Latest (with weighted average)

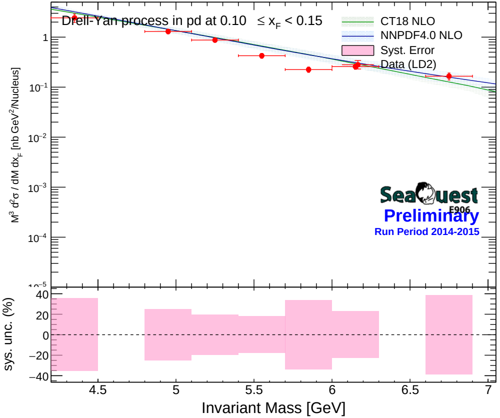


LD2 Cross-Section Previous Vs Latest: $0.10 \leq x_F < 0.15$

Previous

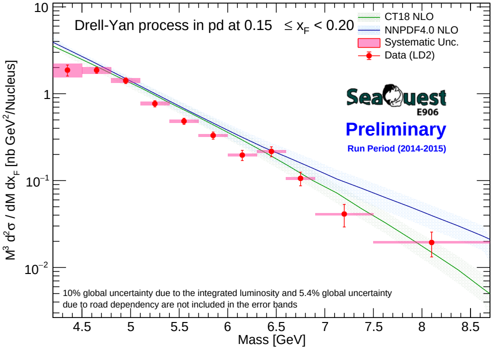


Latest (with weighted average)

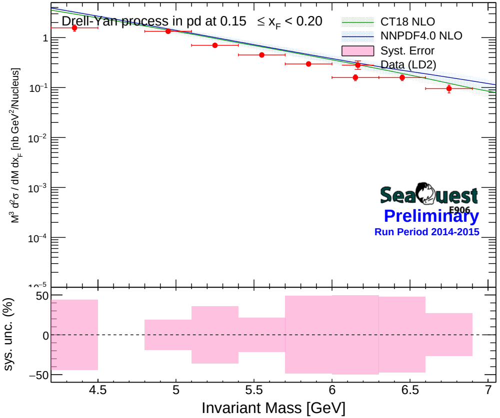


LD2 Cross-Section Previous Vs Latest: $0.15 \leq x_F < 0.20$

Previous

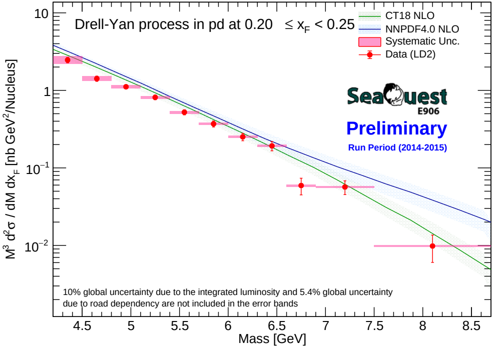


Latest (with weighted average)

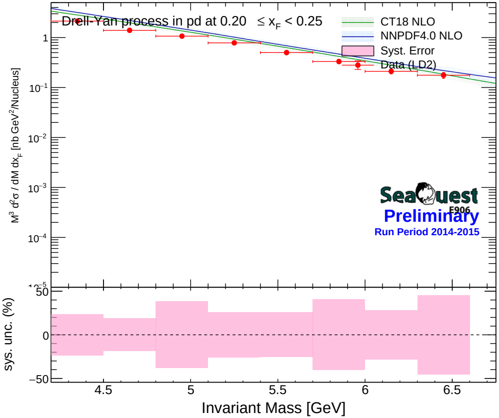


LD2 Cross-Section Previous Vs Latest: $0.20 \leq x_F < 0.25$

Previous

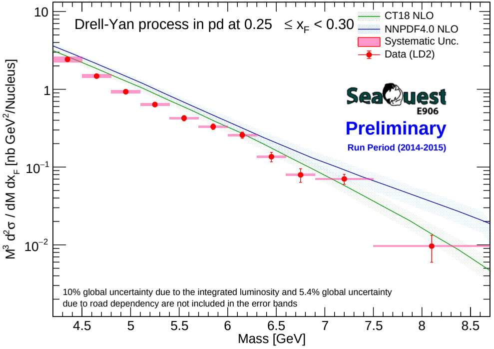


Latest (with weighted average)

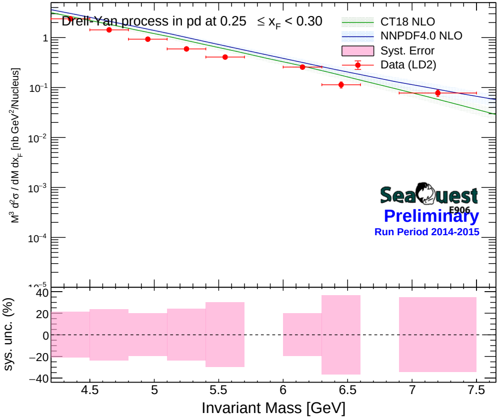


LD2 Cross-Section Previous Vs Latest: $0.25 \leq x_F < 0.30$

Previous

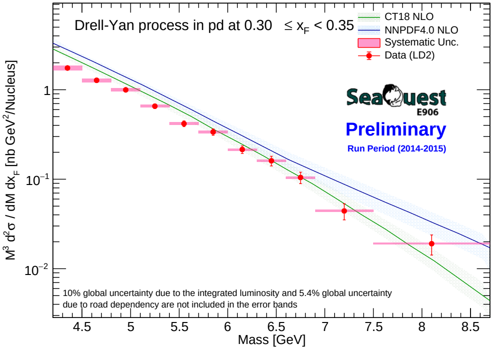


Latest (with weighted average)

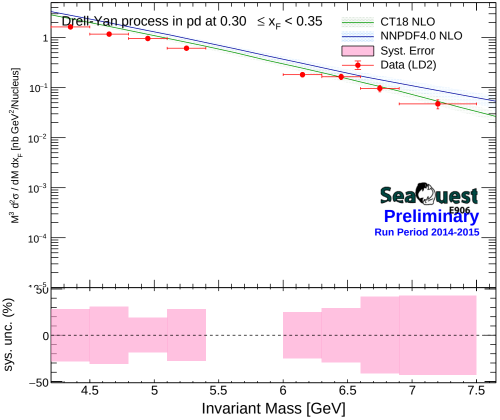


LD2 Cross-Section Previous Vs Latest: $0.30 \le x_F < 0.35$

Previous



Latest (with weighted average)



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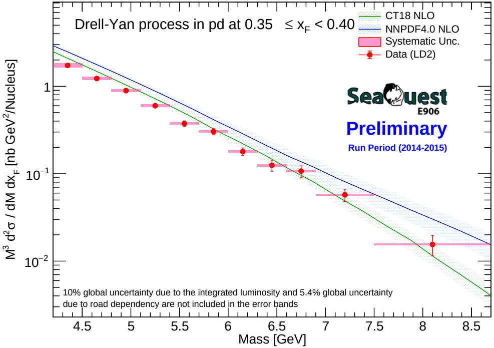
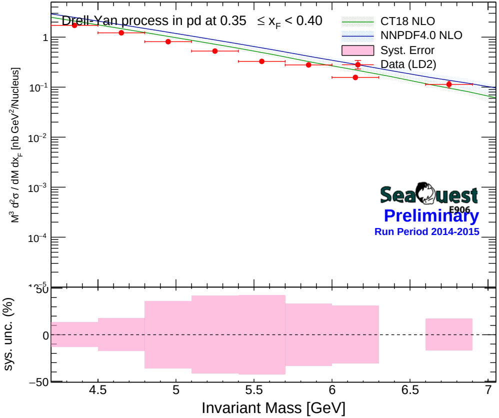
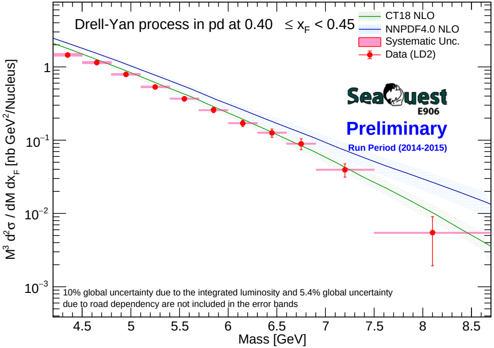


Latest (with weighted average)

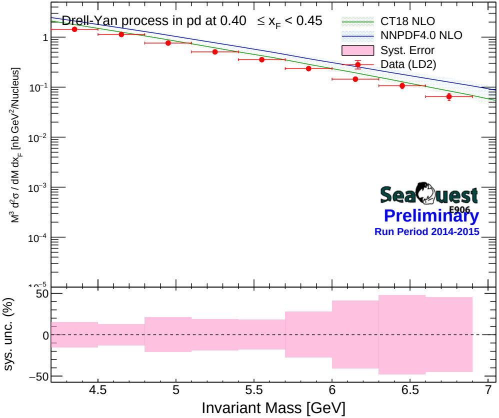


LD2 Cross-Section Previous Vs Latest: $0.40 \leq x_F < 0.45$

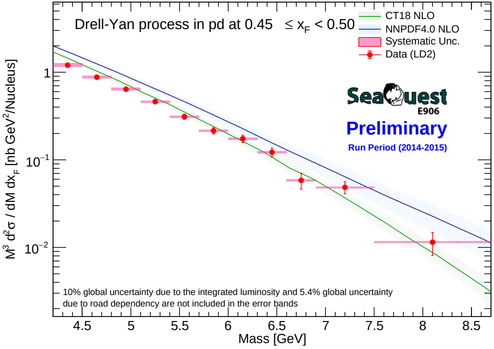
Previous



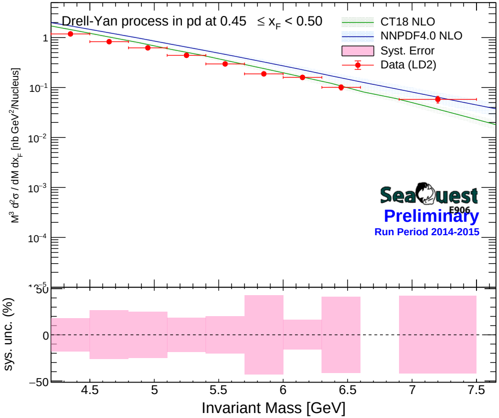
Latest (with weighted average)



Previous

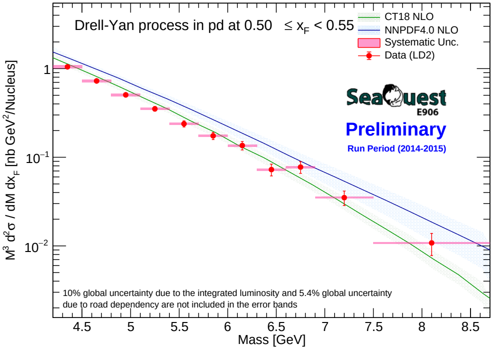


Latest (with weighted average)

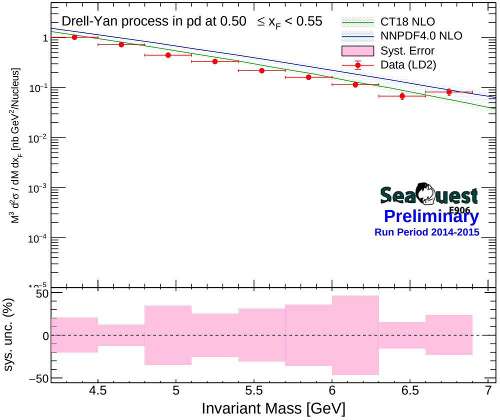


LD2 Cross-Section Previous Vs Latest: $0.50 \leq x_F < 0.55$

Previous

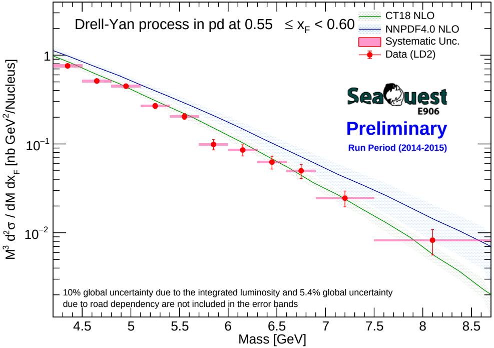


Latest (with weighted average)

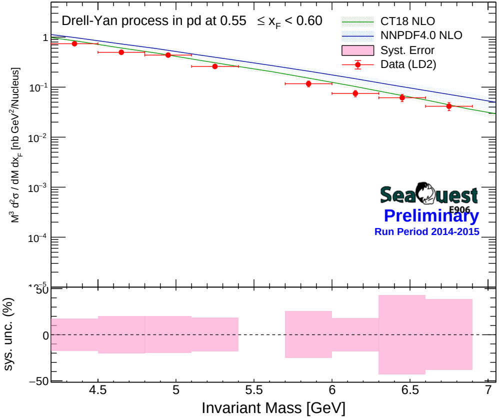


LD2 Cross-Section Previous Vs Latest: $0.55 \leq x_F < 0.60$

Previous

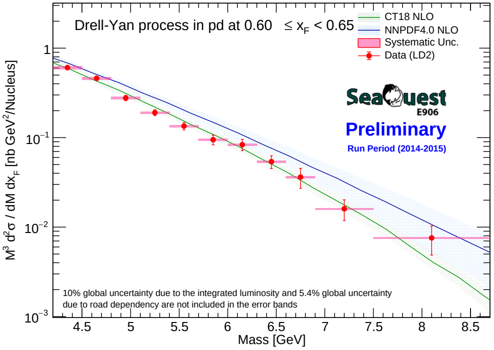


Latest (with weighted average)

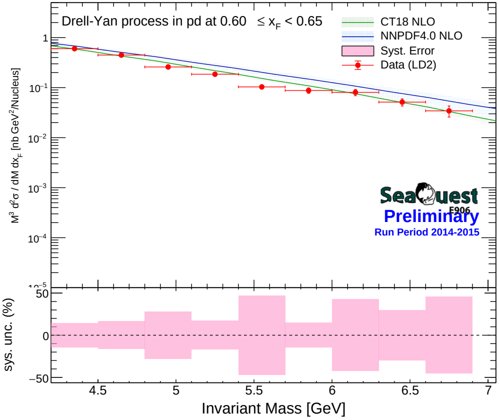


LD2 Cross-Section Previous Vs Latest: $0.60 \leq x_F < 0.65$

Previous

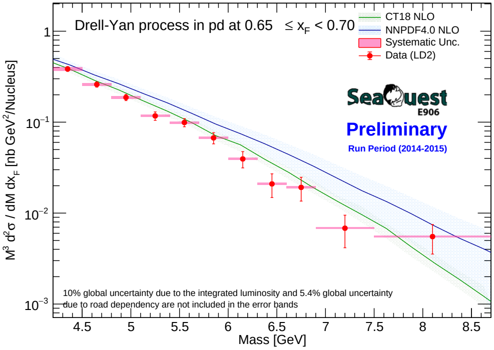


Latest (with weighted average)

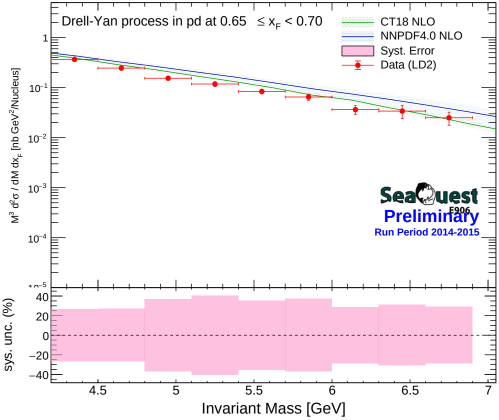


LD2 Cross-Section Previous Vs Latest: $0.65 \leq x_F < 0.70$

Previous



Latest (with weighted average)



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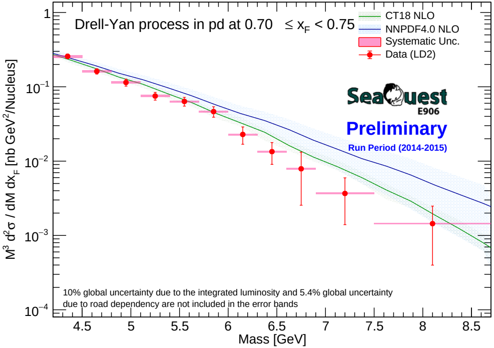
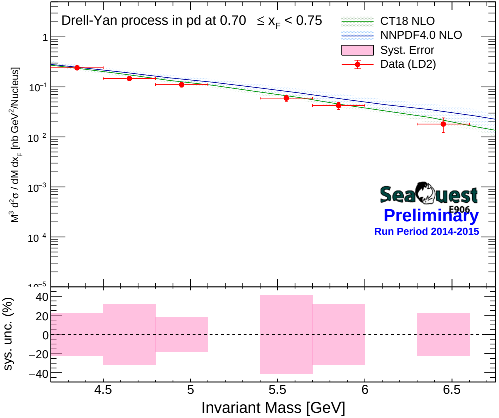
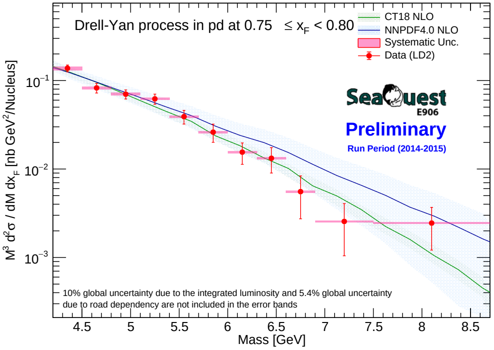


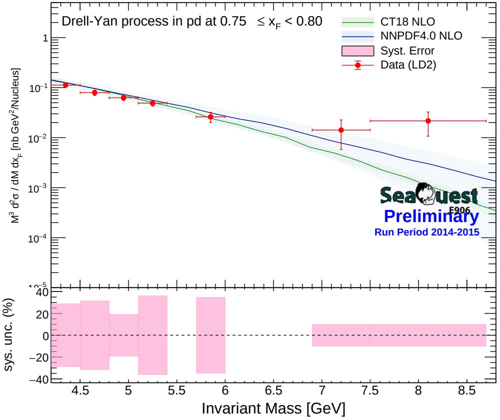
Latest (with weighted average)



Previous

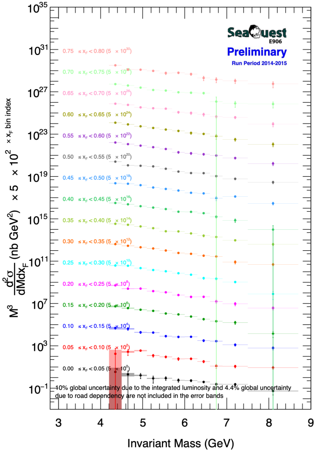


Latest (with weighted average)

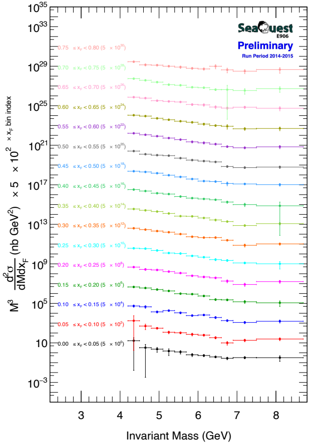


Summary Plots Previous Vs Latest with updated systematics

Previous

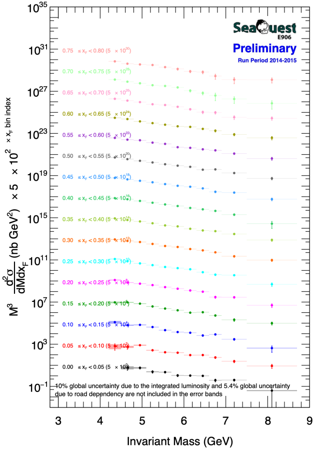


Latest (with weighted average)

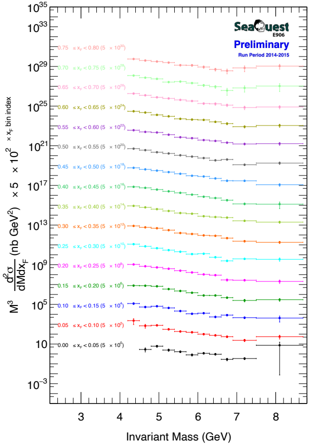


Summary Plots Previous Vs Latest with updated systematics

Previous



Latest (with weighted average)



Backup Slides

Double Differential Systematics (xF and Mass)

Extension to 2D Kinematics:

- The inverse-variance weighting procedure is extended to the double differential bins (xF and Invariant Mass).
- The following slides document the systematic uncertainties strictly for statistically significant data sets across both LH2 and LD2 targets.
- **Filtering Criteria:** Bins reporting a relative variance $\leq 0\%$ or $\geq 50\%$ represent zero-bin artifacts or severe low-statistics starvation and have been omitted for clarity.

Double Differential Systematics (xF and Mass) - LH2 (1/10)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.00, 0.05)$	$[2.20, 2.30)$	4.6647e-01	1.8959e-01	40.64%
$[0.00, 0.05)$	$[2.40, 2.50)$	2.4243e-01	9.2653e-02	38.22%
$[0.00, 0.05)$	$[2.50, 2.60)$	1.2537e-01	5.1962e-02	41.45%
$[0.00, 0.05)$	$[2.60, 2.70)$	1.0592e-01	4.2068e-02	39.72%
$[0.05, 0.10)$	$[2.10, 2.20)$	1.0106e+00	3.5192e-01	34.82%
$[0.05, 0.10)$	$[2.20, 2.30)$	6.1206e-01	2.0965e-01	34.25%
$[0.05, 0.10)$	$[2.30, 2.40)$	2.3946e-01	1.1893e-01	49.67%
$[0.05, 0.10)$	$[2.40, 2.50)$	2.1213e-01	4.1157e-02	19.40%
$[0.05, 0.10)$	$[2.50, 2.60)$	1.4685e-01	5.9468e-02	40.50%
$[0.05, 0.10)$	$[2.60, 2.70)$	1.2250e-01	2.8580e-02	23.33%
$[0.05, 0.10)$	$[2.70, 2.80)$	6.7852e-02	3.2529e-02	47.94%

Double Differential Systematics (xF and Mass) - LH2 (2/10)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.10, 0.15)$	$[2.30, 2.40)$	3.6968e-01	1.0765e-01	29.12%
$[0.10, 0.15)$	$[2.40, 2.50)$	2.2354e-01	4.5725e-02	20.46%
$[0.10, 0.15)$	$[2.60, 2.70)$	1.3294e-01	4.1845e-02	31.48%
$[0.10, 0.15)$	$[2.70, 2.80)$	7.3985e-02	2.4278e-02	32.82%
$[0.15, 0.20)$	$[2.00, 2.10)$	9.1591e-01	4.3322e-01	47.30%
$[0.15, 0.20)$	$[2.10, 2.20)$	7.1875e-01	2.0705e-01	28.81%
$[0.15, 0.20)$	$[2.20, 2.30)$	5.0253e-01	1.2457e-01	24.79%
$[0.15, 0.20)$	$[2.30, 2.40)$	3.6176e-01	1.2212e-01	33.76%
$[0.15, 0.20)$	$[2.40, 2.50)$	1.9252e-01	5.2335e-02	27.18%
$[0.15, 0.20)$	$[2.50, 2.60)$	1.7226e-01	7.6151e-02	44.21%
$[0.15, 0.20)$	$[2.60, 2.70)$	1.1220e-01	2.8566e-02	25.46%

Double Differential Systematics (xF and Mass) - LH2 (3/10)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.20, 0.25)$	$[2.10, 2.20)$	5.4657e-01	2.4949e-01	45.65%
$[0.20, 0.25)$	$[2.20, 2.30)$	4.8145e-01	1.6689e-01	34.66%
$[0.20, 0.25)$	$[2.30, 2.40)$	3.2918e-01	5.4812e-02	16.65%
$[0.20, 0.25)$	$[2.40, 2.50)$	2.3267e-01	3.8954e-02	16.74%
$[0.20, 0.25)$	$[2.60, 2.70)$	9.4423e-02	3.9979e-02	42.34%
$[0.20, 0.25)$	$[2.70, 2.80)$	7.7348e-02	1.0475e-02	13.54%
$[0.20, 0.25)$	$[2.80, 2.90)$	3.6159e-02	5.6716e-03	15.69%
$[0.25, 0.30)$	$[2.00, 2.10)$	7.5260e-01	1.2825e-01	17.04%
$[0.25, 0.30)$	$[2.10, 2.20)$	6.9524e-01	1.9496e-01	28.04%
$[0.25, 0.30)$	$[2.20, 2.30)$	4.5304e-01	5.9077e-02	13.04%
$[0.25, 0.30)$	$[2.30, 2.40)$	2.4852e-01	8.9635e-02	36.07%

Double Differential Systematics (xF and Mass) - LH2 (4/10)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.25, 0.30)$	$[2.50, 2.60)$	1.2740e-01	4.2815e-02	33.61%
$[0.25, 0.30)$	$[2.60, 2.70)$	9.4024e-02	1.5640e-02	16.63%
$[0.25, 0.30)$	$[2.70, 2.80)$	5.3786e-02	1.6526e-02	30.72%
$[0.25, 0.30)$	$[2.80, 2.90)$	4.4295e-02	1.9575e-02	44.19%
$[0.25, 0.30)$	$[2.90, 3.00)$	2.4688e-02	9.9960e-03	40.49%
$[0.30, 0.35)$	$[2.00, 2.10)$	7.9903e-01	2.3076e-01	28.88%
$[0.30, 0.35)$	$[2.10, 2.20)$	5.8222e-01	1.2269e-01	21.07%
$[0.30, 0.35)$	$[2.20, 2.30)$	4.1542e-01	6.2699e-02	15.09%
$[0.30, 0.35)$	$[2.30, 2.40)$	3.1708e-01	4.5316e-02	14.29%
$[0.30, 0.35)$	$[2.40, 2.50)$	2.0555e-01	3.2558e-02	15.84%
$[0.30, 0.35)$	$[2.50, 2.60)$	1.1104e-01	3.6713e-02	33.06%

Double Differential Systematics (xF and Mass) - LH2 (5/10)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.30, 0.35)$	$[2.70, 2.80)$	9.1170e-02	1.2310e-02	13.50%
$[0.30, 0.35)$	$[2.80, 2.90)$	4.8748e-02	9.6601e-03	19.82%
$[0.35, 0.40)$	$[2.00, 2.10)$	7.0876e-01	1.0139e-01	14.31%
$[0.35, 0.40)$	$[2.10, 2.20)$	5.2199e-01	1.5482e-01	29.66%
$[0.35, 0.40)$	$[2.20, 2.30)$	3.9245e-01	6.0602e-02	15.44%
$[0.35, 0.40)$	$[2.30, 2.40)$	2.5106e-01	4.9200e-02	19.60%
$[0.35, 0.40)$	$[2.40, 2.50)$	1.3409e-01	2.9509e-02	22.01%
$[0.35, 0.40)$	$[2.50, 2.60)$	1.2855e-01	3.2383e-02	25.19%
$[0.35, 0.40)$	$[2.60, 2.70)$	8.8306e-02	1.4040e-02	15.90%
$[0.35, 0.40)$	$[2.70, 2.80)$	5.5295e-02	2.4767e-02	44.79%
$[0.35, 0.40)$	$[2.80, 2.90)$	3.8024e-02	1.6029e-02	42.16%

Double Differential Systematics (xF and Mass) - LH2 (6/10)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.40, 0.45)$	$[2.10, 2.20)$	4.6687e-01	1.0521e-01	22.54%
$[0.40, 0.45)$	$[2.20, 2.30)$	2.8022e-01	9.1991e-02	32.83%
$[0.40, 0.45)$	$[2.30, 2.40)$	2.3071e-01	3.3997e-02	14.74%
$[0.40, 0.45)$	$[2.50, 2.60)$	8.9501e-02	2.7328e-02	30.53%
$[0.40, 0.45)$	$[2.60, 2.70)$	7.4179e-02	2.9933e-02	40.35%
$[0.40, 0.45)$	$[2.70, 2.80)$	6.1148e-02	1.5443e-02	25.26%
$[0.40, 0.45)$	$[2.80, 2.90)$	3.5909e-02	1.1889e-02	33.11%
$[0.45, 0.50)$	$[2.00, 2.10)$	4.6946e-01	8.7180e-02	18.57%
$[0.45, 0.50)$	$[2.10, 2.20)$	4.0052e-01	9.1889e-02	22.94%
$[0.45, 0.50)$	$[2.20, 2.30)$	2.9739e-01	4.2426e-02	14.27%
$[0.45, 0.50)$	$[2.30, 2.40)$	1.8520e-01	2.3280e-02	12.57%

Double Differential Systematics (xF and Mass) - LH2 (7/10)

xF Range	**Mass Range [GeV]**	**Mean σ_w**	**Std Dev $\sigma_{sys,w}$**	**Relative Error**
$[0.45, 0.50)$	$[2.50, 2.60)$	7.6792e-02	3.5737e-02	46.54%
$[0.45, 0.50)$	$[2.60, 2.70)$	4.7282e-02	1.9263e-02	40.74%
$[0.50, 0.55)$	$[2.00, 2.10)$	5.1561e-01	4.3113e-02	8.36%
$[0.50, 0.55)$	$[2.10, 2.20)$	2.4825e-01	3.8514e-02	15.51%
$[0.50, 0.55)$	$[2.20, 2.30)$	1.9861e-01	2.6949e-02	13.57%
$[0.50, 0.55)$	$[2.30, 2.40)$	1.7728e-01	5.6211e-02	31.71%
$[0.50, 0.55)$	$[2.40, 2.50)$	9.9353e-02	2.3469e-02	23.62%
$[0.50, 0.55)$	$[2.50, 2.60)$	6.3380e-02	1.6129e-02	25.45%
$[0.50, 0.55)$	$[2.60, 2.70)$	5.0313e-02	1.0344e-02	20.56%
$[0.50, 0.55)$	$[2.70, 2.80)$	4.1483e-02	9.6799e-03	23.33%
$[0.55, 0.60)$	$[2.00, 2.10)$	3.2536e-01	1.0254e-02	3.15%

Double Differential Systematics (xF and Mass) - LH2 (8/10)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.55, 0.60)$	$[2.20, 2.30)$	1.6549e-01	3.3759e-02	20.40%
$[0.55, 0.60)$	$[2.30, 2.40)$	1.0744e-01	3.5247e-02	32.81%
$[0.55, 0.60)$	$[2.40, 2.50)$	7.4276e-02	1.4861e-02	20.01%
$[0.55, 0.60)$	$[2.50, 2.60)$	4.4520e-02	1.0677e-02	23.98%
$[0.55, 0.60)$	$[2.60, 2.70)$	3.5292e-02	1.6495e-02	46.74%
$[0.55, 0.60)$	$[2.70, 2.80)$	2.6122e-02	6.3284e-03	24.23%
$[0.60, 0.65)$	$[2.00, 2.10)$	2.1866e-01	8.9135e-02	40.76%
$[0.60, 0.65)$	$[2.10, 2.20)$	1.7900e-01	2.4785e-02	13.85%
$[0.60, 0.65)$	$[2.20, 2.30)$	1.0609e-01	1.6341e-02	15.40%
$[0.60, 0.65)$	$[2.30, 2.40)$	7.6175e-02	1.6733e-02	21.97%
$[0.60, 0.65)$	$[2.40, 2.50)$	5.4076e-02	1.3899e-02	25.70%

Double Differential Systematics (xF and Mass) - LH2 (9/10)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.60, 0.65)$	$[2.60, 2.70)$	2.7941e-02	1.3177e-02	47.16%
$[0.65, 0.70)$	$[2.00, 2.10)$	1.4650e-01	1.7599e-02	12.01%
$[0.65, 0.70)$	$[2.10, 2.20)$	1.1789e-01	1.4840e-02	12.59%
$[0.65, 0.70)$	$[2.20, 2.30)$	6.4554e-02	1.7128e-02	26.53%
$[0.65, 0.70)$	$[2.30, 2.40)$	5.4773e-02	2.1213e-02	38.73%
$[0.65, 0.70)$	$[2.40, 2.50)$	4.8254e-02	8.8236e-03	18.29%
$[0.65, 0.70)$	$[2.50, 2.60)$	2.6751e-02	1.1587e-02	43.31%
$[0.65, 0.70)$	$[2.60, 2.70)$	2.4311e-02	5.1266e-03	21.09%
$[0.70, 0.75)$	$[2.00, 2.10)$	1.1002e-01	1.0678e-02	9.71%
$[0.70, 0.75)$	$[2.10, 2.20)$	7.4866e-02	2.2381e-02	29.89%
$[0.70, 0.75)$	$[2.20, 2.30)$	4.8945e-02	1.2593e-02	25.73%

Double Differential Systematics (xF and Mass) - LH2 (10/10)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.70, 0.75)$	$[2.40, 2.50)$	2.3570e-02	8.6238e-03	36.59%
$[0.70, 0.75)$	$[2.60, 2.70)$	1.3325e-02	5.8748e-03	44.09%
$[0.75, 0.80)$	$[2.00, 2.10)$	5.6607e-02	1.2918e-02	22.82%
$[0.75, 0.80)$	$[2.20, 2.30)$	2.5546e-02	1.0505e-02	41.12%
$[0.75, 0.80)$	$[2.30, 2.40)$	2.0004e-02	8.5603e-03	42.79%
$[0.75, 0.80)$	$[2.40, 2.50)$	1.6363e-02	6.0294e-03	36.85%
$[0.75, 0.80)$	$[2.50, 2.60)$	1.2196e-02	6.0025e-03	49.21%

Double Differential Systematics (xF and Mass) - LD2 (1/11)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.00, 0.05)$	$[2.20, 2.30)$	1.2134e+00	3.5467e-01	29.23%
$[0.00, 0.05)$	$[2.30, 2.40)$	4.9300e-01	2.2986e-01	46.62%
$[0.00, 0.05)$	$[2.60, 2.70)$	2.3350e-01	7.9457e-02	34.03%
$[0.00, 0.05)$	$[2.70, 2.80)$	1.8693e-01	5.7193e-02	30.60%
$[0.05, 0.10)$	$[2.20, 2.30)$	1.5352e+00	2.7104e-01	17.66%
$[0.05, 0.10)$	$[2.30, 2.40)$	6.1204e-01	1.1169e-01	18.25%
$[0.05, 0.10)$	$[2.40, 2.50)$	3.6952e-01	1.6648e-01	45.05%
$[0.05, 0.10)$	$[2.50, 2.60)$	3.0688e-01	1.1042e-01	35.98%
$[0.05, 0.10)$	$[2.60, 2.70)$	1.9908e-01	3.1577e-02	15.86%
$[0.05, 0.10)$	$[2.80, 2.90)$	1.0977e-01	4.3818e-02	39.92%
$[0.10, 0.15)$	$[2.00, 2.10)$	2.4320e+00	4.6782e-01	19.24%

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.10, 0.15)$	$[2.30, 2.40)$	8.7415e-01	1.3958e-01	15.97%
$[0.10, 0.15)$	$[2.40, 2.50)$	4.2530e-01	5.9980e-02	14.10%
$[0.10, 0.15)$	$[2.50, 2.60)$	2.2473e-01	7.1942e-02	32.01%
$[0.10, 0.15)$	$[2.60, 2.70)$	2.5785e-01	5.2115e-02	20.21%
$[0.10, 0.15)$	$[2.80, 2.90)$	1.6552e-01	6.1436e-02	37.12%
$[0.15, 0.20)$	$[2.00, 2.10)$	1.5687e+00	6.1198e-01	39.01%
$[0.15, 0.20)$	$[2.10, 2.20)$	1.5232e+00	7.4036e-01	48.60%
$[0.15, 0.20)$	$[2.20, 2.30)$	1.3359e+00	1.9766e-01	14.80%
$[0.15, 0.20)$	$[2.30, 2.40)$	7.0016e-01	2.3754e-01	33.93%
$[0.15, 0.20)$	$[2.40, 2.50)$	4.4981e-01	8.3544e-02	18.57%
$[0.15, 0.20)$	$[2.50, 2.60)$	2.9692e-01	1.4121e-01	47.56%
$[0.15, 0.20)$	$[2.60, 2.70)$	[illegible]	[illegible]	[illegible]

Double Differential Systematics (xF and Mass) - LD2 (3/11)

xF Range	**Mass Range [GeV]**	**Mean σ_w**	**Std Dev $\sigma_{sys,w}$**	**Relative Error**
$[0.15, 0.20)$	$[2.70, 2.80)$	1.5843e-01	7.3299e-02	46.27%
$[0.15, 0.20)$	$[2.80, 2.90)$	9.5246e-02	2.3443e-02	24.61%
$[0.20, 0.25)$	$[2.00, 2.10)$	2.1372e+00	3.8561e-01	18.04%
$[0.20, 0.25)$	$[2.10, 2.20)$	1.3985e+00	1.9926e-01	14.25%
$[0.20, 0.25)$	$[2.20, 2.30)$	1.0683e+00	3.9122e-01	36.62%
$[0.20, 0.25)$	$[2.30, 2.40)$	7.8453e-01	1.8395e-01	23.45%
$[0.20, 0.25)$	$[2.40, 2.50)$	5.0149e-01	1.1740e-01	23.41%
$[0.20, 0.25)$	$[2.50, 2.60)$	3.3169e-01	1.2986e-01	39.15%
$[0.20, 0.25)$	$[2.60, 2.70)$	2.1104e-01	5.5104e-02	26.11%
$[0.20, 0.25)$	$[2.70, 2.80)$	1.7673e-01	7.7991e-02	44.13%
$[0.20, 0.25)$	$[2.80, 2.90)$	5.8350e-02	2.8937e-02	49.59%

Double Differential Systematics (xF and Mass) - LD2 (4/11)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.25, 0.30)$	$[2.10, 2.20)$	1.4279e+00	2.9626e-01	20.75%
$[0.25, 0.30)$	$[2.20, 2.30)$	9.2956e-01	1.5364e-01	16.53%
$[0.25, 0.30)$	$[2.30, 2.40)$	5.9333e-01	1.2729e-01	21.45%
$[0.25, 0.30)$	$[2.40, 2.50)$	4.0752e-01	1.1414e-01	28.01%
$[0.25, 0.30)$	$[2.60, 2.70)$	2.5680e-01	4.2894e-02	16.70%
$[0.25, 0.30)$	$[2.70, 2.80)$	1.1377e-01	3.9897e-02	35.07%
$[0.25, 0.30)$	$[2.90, 3.00)$	7.7665e-02	2.5707e-02	33.10%
$[0.30, 0.35)$	$[2.00, 2.10)$	1.6374e+00	4.2036e-01	25.67%
$[0.30, 0.35)$	$[2.10, 2.20)$	1.1762e+00	3.4070e-01	28.97%
$[0.30, 0.35)$	$[2.20, 2.30)$	9.6178e-01	1.4852e-01	15.44%
$[0.30, 0.35)$	$[2.30, 2.40)$	6.1483e-01	1.5873e-01	25.82%

Double Differential Systematics (xF and Mass) - LD2 (5/11)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.30, 0.35)$	$[2.70, 2.80)$	1.6502e-01	4.4954e-02	27.24%
$[0.30, 0.35)$	$[2.80, 2.90)$	9.6576e-02	3.8679e-02	40.05%
$[0.30, 0.35)$	$[2.90, 3.00)$	4.7350e-02	1.9646e-02	41.49%
$[0.35, 0.40)$	$[2.00, 2.10)$	1.7182e+00	1.1746e-01	6.84%
$[0.35, 0.40)$	$[2.10, 2.20)$	1.2202e+00	1.6988e-01	13.92%
$[0.35, 0.40)$	$[2.20, 2.30)$	8.1179e-01	2.7942e-01	34.42%
$[0.35, 0.40)$	$[2.30, 2.40)$	5.2692e-01	2.1254e-01	40.34%
$[0.35, 0.40)$	$[2.40, 2.50)$	3.2734e-01	1.3486e-01	41.20%
$[0.35, 0.40)$	$[2.50, 2.60)$	2.7822e-01	8.7961e-02	31.62%
$[0.35, 0.40)$	$[2.60, 2.70)$	1.5650e-01	4.5772e-02	29.25%
$[0.35, 0.40)$	$[2.80, 2.90)$	1.1346e-01	1.5148e-02	13.35%

Double Differential Systematics (xF and Mass) - LD2 (6/11)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.40, 0.45)$	$[2.10, 2.20)$	1.1311e+00	7.9911e-02	7.07%
$[0.40, 0.45)$	$[2.20, 2.30)$	7.6144e-01	1.3688e-01	17.98%
$[0.40, 0.45)$	$[2.30, 2.40)$	5.0685e-01	7.9701e-02	15.72%
$[0.40, 0.45)$	$[2.40, 2.50)$	3.5430e-01	5.2289e-02	14.76%
$[0.40, 0.45)$	$[2.50, 2.60)$	2.3511e-01	6.0446e-02	25.71%
$[0.40, 0.45)$	$[2.60, 2.70)$	1.4468e-01	5.7135e-02	39.49%
$[0.40, 0.45)$	$[2.70, 2.80)$	1.0666e-01	4.9775e-02	46.67%
$[0.40, 0.45)$	$[2.80, 2.90)$	6.4523e-02	2.8254e-02	43.79%
$[0.45, 0.50)$	$[2.00, 2.10)$	1.1839e+00	1.6772e-01	14.17%
$[0.45, 0.50)$	$[2.10, 2.20)$	8.2842e-01	2.0026e-01	24.17%
$[0.45, 0.50)$	$[2.20, 2.30)$	6.2343e-01	1.4134e-01	22.67%
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Double Differential Systematics (xF and Mass) - LD2 (7/11)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.45, 0.50)$	$[2.40, 2.50)$	2.9707e-01	5.0997e-02	17.17%
$[0.45, 0.50)$	$[2.50, 2.60)$	1.8791e-01	7.8155e-02	41.59%
$[0.45, 0.50)$	$[2.60, 2.70)$	1.5975e-01	1.9869e-02	12.44%
$[0.45, 0.50)$	$[2.70, 2.80)$	1.0093e-01	4.0190e-02	39.82%
$[0.45, 0.50)$	$[2.90, 3.00)$	5.7965e-02	2.3706e-02	40.90%
$[0.50, 0.55)$	$[2.00, 2.10)$	1.0189e+00	1.7374e-01	17.05%
$[0.50, 0.55)$	$[2.10, 2.20)$	7.2468e-01	4.5340e-02	6.26%
$[0.50, 0.55)$	$[2.20, 2.30)$	4.4702e-01	1.4831e-01	33.18%
$[0.50, 0.55)$	$[2.30, 2.40)$	3.3354e-01	7.6521e-02	22.94%
$[0.50, 0.55)$	$[2.40, 2.50)$	2.1909e-01	6.4012e-02	29.22%
$[0.50, 0.55)$	$[2.50, 2.60)$	1.6166e-01	5.5508e-02	34.34%
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Double Differential Systematics (xF and Mass) - LD2 (8/11)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.50, 0.55)$	$[2.70, 2.80)$	6.7888e-02	7.7766e-03	11.45%
$[0.50, 0.55)$	$[2.80, 2.90)$	8.2012e-02	1.7043e-02	20.78%
$[0.55, 0.60)$	$[2.00, 2.10)$	7.3861e-01	1.0244e-01	13.87%
$[0.55, 0.60)$	$[2.10, 2.20)$	4.9734e-01	8.5445e-02	17.18%
$[0.55, 0.60)$	$[2.20, 2.30)$	4.3767e-01	7.4056e-02	16.92%
$[0.55, 0.60)$	$[2.30, 2.40)$	2.5958e-01	3.8744e-02	14.93%
$[0.55, 0.60)$	$[2.50, 2.60)$	1.1703e-01	2.7149e-02	23.20%
$[0.55, 0.60)$	$[2.60, 2.70)$	7.4815e-02	1.1092e-02	14.83%
$[0.55, 0.60)$	$[2.70, 2.80)$	6.1416e-02	2.5615e-02	41.71%
$[0.55, 0.60)$	$[2.80, 2.90)$	4.1506e-02	1.5355e-02	37.00%
$[0.60, 0.65)$	$[2.00, 2.10)$	6.0184e-01	5.6361e-02	9.36%

Double Differential Systematics (xF and Mass) - LD2 (9/11)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.60, 0.65)$	$[2.20, 2.30)$	2.5929e-01	6.7410e-02	26.00%
$[0.60, 0.65)$	$[2.30, 2.40)$	1.8560e-01	2.5284e-02	13.62%
$[0.60, 0.65)$	$[2.40, 2.50)$	1.0353e-01	4.7449e-02	45.83%
$[0.60, 0.65)$	$[2.50, 2.60)$	8.7706e-02	8.9969e-03	10.26%
$[0.60, 0.65)$	$[2.60, 2.70)$	8.0616e-02	3.3267e-02	41.27%
$[0.60, 0.65)$	$[2.70, 2.80)$	5.1333e-02	1.4382e-02	28.02%
$[0.60, 0.65)$	$[2.80, 2.90)$	3.4400e-02	1.5218e-02	44.24%
$[0.65, 0.70)$	$[2.00, 2.10)$	3.6757e-01	8.9009e-02	24.22%
$[0.65, 0.70)$	$[2.10, 2.20)$	2.4621e-01	6.0907e-02	24.74%
$[0.65, 0.70)$	$[2.20, 2.30)$	1.5429e-01	5.4661e-02	35.43%
$[0.65, 0.70)$	$[2.30, 2.40)$	1.1833e-01	4.6189e-02	39.03%

Double Differential Systematics (xF and Mass) - LD2 (10/11)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.65, 0.70)$	$[2.50, 2.60)$	6.5078e-02	2.3231e-02	35.70%
$[0.65, 0.70)$	$[2.60, 2.70)$	3.6397e-02	9.6883e-03	26.62%
$[0.65, 0.70)$	$[2.70, 2.80)$	3.3985e-02	9.8990e-03	29.13%
$[0.65, 0.70)$	$[2.80, 2.90)$	2.5011e-02	6.7470e-03	26.98%
$[0.70, 0.75)$	$[2.00, 2.10)$	2.4122e-01	4.5913e-02	19.03%
$[0.70, 0.75)$	$[2.10, 2.20)$	1.4718e-01	4.3820e-02	29.77%
$[0.70, 0.75)$	$[2.20, 2.30)$	1.1097e-01	1.6766e-02	15.11%
$[0.70, 0.75)$	$[2.40, 2.50)$	5.9655e-02	2.3853e-02	39.99%
$[0.70, 0.75)$	$[2.50, 2.60)$	4.2258e-02	1.2634e-02	29.90%
$[0.70, 0.75)$	$[2.70, 2.80)$	1.8082e-02	3.5747e-03	19.77%
$[0.75, 0.80)$	$[2.00, 2.10)$	1.1242e-01	3.0062e-02	26.74%

Double Differential Systematics (xF and Mass) - LD2 (11/11)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
$[0.75, 0.80)$	$[2.20, 2.30)$	6.2737e-02	1.0170e-02	16.21%
$[0.75, 0.80)$	$[2.30, 2.40)$	4.9022e-02	1.7038e-02	34.76%
$[0.75, 0.80)$	$[2.50, 2.60)$	2.6054e-02	8.7078e-03	33.42%

Discussion & Next Steps

Why is the relative error still $\sim$13–18%?

1. **Statistical Noise Limitations:** While inverse-variance weighting heavily suppresses the noise of low-POT datasets (RS59, RS70), it cannot completely eliminate the pull from extreme outliers.
2. **Global Acceptance Mismatch:** A single `acceptance_mass_xF.root` file is currently used across all roadsets. Because different triggers have distinct kinematic acceptances, dividing by a global average creates artificial variance.

Path Forward: Implement roadset-specific Monte Carlo acceptance maps and evaluate the necessity of excluding extremely low-POT datasets from the overall variance calculation.